



education

phd | machine learning

uc berkeley | '17-'22

research: interpretable ml

advisor: bin yu

bs | cs & math

university of virginia | '14-'17

double major

skills

language models | deep learning

data science | data cleaning

huggingface | pytorch

rule-based models | causal inference

awards

berkeley grad slam semifinalist '19, '22

pdsoros fellowship finalist '19

outstanding teaching award '18

uva rader research award '17

uva undergrad symposium winner '17

raven honor society '16-'17

icpc regional qualification '14-'16

1st place microsoft code jam '16

3rd place google games uva '17

2nd place apt puzzle competition '17

rodman scholarship '14-'17

teaching

berkeley | fall 2019

artificial intelligence: cs 188

berkeley | summer 2018

machine learning: cs 189/289

lectures to class of 80+ students

service

volunteering

basis education volunteering '19-'22

bair undergrad mentoring '18-'22

computer literacy volunteering '15-'17

organizer

iclr trustml workshop '26

area chair

ml4h '25 | xxai workshop, ml4h '24

reviewer

iclr, icml '26

arr, iclr, icml, iccv, neurips, pnas '25

iclr, icml, neurips '24

neurips '23 | acl '22

iclr, cvpr, aaai, neurips '21 | neurips '20

experience

microsoft research

senior researcher (deep learning group) | summer '22 - present

- improving the interpretability of llms
- researching knowledge discovery with llms
- building next-generation foundation models

health tech

paige ai | research scientist | summer '21 - summer '22

- interpretable deep learning in digital pathology (especially bladder cancer)

response4life | volunteer data scientist | spring '20

- helped develop, integrate, and deploy models to forecast covid-19 severity

pacmed ai | healthcare ml intern | summer '19

- developed interpretable, tabular ml models for healthcare

phd

berkeley | interpretable ml research (bin yu group) | fall '17 - spring '22

- developed post-hoc interpretation methods for ml models (e.g. neural nets)
- developed interpretable models in medicine, biology, and computer vision

aws | ml fairness intern (pietro perona group) | summer '20

- testing for bias with causal matching using GANs

meta ai | computer vision intern | summer '17

- investigated unsupervised deep learning for segmentation of satellite imagery

undergrad

hhmi | ml research (srini turaga group) | summer '14, '15, '16

- researched neural image segmentation and biophysical simulations

uva | ml research (yanjun qi group) | fall '16 - spring '17

- developed multi-task graphical models for analyzing functional brain connectivity

uva | comp. neuroscience research (william levy group) | fall '14 - fall '16

- developed biophysical models of single-neuron computation

selected publications

interpretability $\times$ llms $\rightarrow$ science

- augmenting interpretable models with llms: cs, et al. *nature comm.*, '23
- generative causal testing with llms: antonello*, cs*, et al. *nature neuroscience*, '24
- interpretable embeddings by asking llms questions benara*, cs*, et al. *neurips*, '24

interpretability $\times$ rules $\rightarrow$ healthcare

- human-ai co-design for clinical models: feng*, kothari*, ..., cs. *npj digital med* '26
- fast interpretable greedy-tree sums: tan*, cs*, nasseri*, agarwal* et al. *pnas* '22
- imodels: an interpretability package: cs*, nasseri*, tan, tang, & yu. *joss* '21

interpretability $\times$ deep learning $\rightarrow$ general domain

- post-hoc attention steering for llms: zhang, cs, et al. *iclr* '24
- adaptive wavelet distillation from dnns: ha, cs, et al. *neurips* '21
- hierarchical interpretations for dnn predictions: cs*, murdoch*, & yu. *iclr* '19